

Chapter 12

§12.1.

Distance Formula

$$P_1(x_1, y_1, z_1), P_2(x_2, y_2, z_2)$$

$$|P_1 P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Equation of a sphere centered at (h, k, ℓ) with radius r .

$$(x-h)^2 + (y-k)^2 + (z-\ell)^2 = r^2$$

§12.2.

Vector addition $\rightarrow$ the triangle law, the parallelogram law.

Scalar multiplication of a vector

Length or magnitude:

$$\vec{i} = \langle 1, 0, 0 \rangle$$

$$a = \langle a_1, a_2, a_3 \rangle \text{ or } a = a_1 \vec{i} + a_2 \vec{j} + a_3 \vec{k} \quad \vec{j} = \langle 0, 1, 0 \rangle$$

$$|a| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

Properties of vector operations.

unit vector $\rightarrow$ a vector whose magnitude is 1.

Given a vector $a \neq 0$, the unit vector that has the same direction as a is

$$u = \frac{a}{|a|}$$

§12.3.

dot-product: $a = \langle a_1, a_2, a_3 \rangle, b = \langle b_1, b_2, b_3 \rangle$

$$a \cdot b = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Properties of the dot-product

$$a \cdot a = |a|^2, \quad a \cdot b = b \cdot a, \quad a(b+c) = a \cdot b + a \cdot c, \quad (\alpha a) \cdot b = \alpha(a \cdot b) = a \cdot (\alpha b).$$

The Cosine law

Let θ be the angle between vectors a and b , then

$$\cos \theta = \frac{a \cdot b}{|a||b|} \quad \text{or} \quad a \cdot b = |a||b| \cos \theta$$

i) If $a \cdot b > 0$, then $0 \leq \theta < \frac{\pi}{2}$

ii) If $a \cdot b = 0$, then $\theta = \frac{\pi}{2}$, i.e. a is orthogonal to b .

iii) If $a \cdot b < 0$, then $\frac{\pi}{2} < \theta \leq \pi$.

directional angles

$$a = \langle a_1, a_2, a_3 \rangle$$

$$\cos \alpha = \frac{a_1}{|a|}$$

$$\cos \beta = \frac{a_2}{|a|}$$

$$\cos \gamma = \frac{a_3}{|a|}$$

Projection.

$$\text{Proj}_a b = \left(\frac{a \cdot b}{|a|^2} \right) \cdot \frac{a}{|a|} = \frac{a \cdot b}{|a|^2} a \leftarrow \text{vector projection of } b \text{ onto } a$$

$$\text{Comp}_a b = \frac{a \cdot b}{|a|} \leftarrow \text{scalar projection.}$$

§12.4.

Cross-product: Let $a = \langle a_1, a_2, a_3 \rangle$, $b = \langle b_1, b_2, b_3 \rangle$

$$a \times b = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \hat{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \hat{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \hat{k}$$

$$= (a_2 b_3 - a_3 b_2) \hat{i} - (a_1 b_3 - a_3 b_1) \hat{j} + (a_1 b_2 - a_2 b_1) \hat{k}.$$

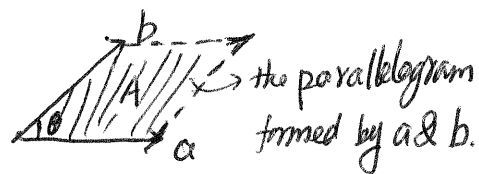
Remark: i) $a \times a = 0$.

ii) $a \times b$ is orthogonal to both a and b .

The Sine Law

p3

$$|a \times b| = |a||b| \sin \theta = \text{Area}(A) \Rightarrow$$



Remark: $a \times b = 0$ iff a and b are parallel.

Properties of the cross-product:

$$a \times b = -b \times a, \quad (\alpha a) \times b = \alpha(a \times b) = a \times (\alpha b), \quad a \times (b + c) = a \times b + a \times c,$$

$$(a + b) \times c = a \times c + b \times c, \quad a \cdot (b \times c) = (a \times b) \cdot c, \quad a \times (b \times c) = (a \cdot c)b - (a \cdot b)c.$$

Triple product

$$a \cdot (b \times c) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

§12.5.

Line L : passing a point $P_0(x_0, y_0, z_0)$, direction $V = \langle a, b, c \rangle$.

Vector equation: $\langle x, y, z \rangle = \langle x_0 + ta, y_0 + tb, z_0 + tc \rangle$

parametric equations: $x = x_0 + at, y = y_0 + bt, z = z_0 + ct$

symmetric equations: $\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$.

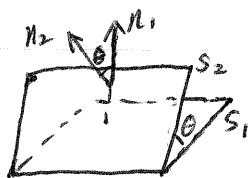
Relation of two lines:

i) Parallel ii) Intersected iii) Skew (neither parallel or intersected).

Plane S : passing a point $P_0(x_0, y_0, z_0)$, normal vector $n = \langle a, b, c \rangle$

Scalar equation: $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$.

Angle between planes



$$\cos \theta = \frac{|n_1 \cdot n_2|}{|n_1||n_2|} \quad (0 \leq \theta \leq \frac{\pi}{2})$$

where n_1 is a normal vector of S_1 .

n_2 is a normal vector of S_2 .

Quadratic surfaces

general form: $AX^2 + BY^2 + CZ^2 + DXY + EYZ + FXZ + GX + HY + IZ + J = 0$.

by translation and rotation, it can be converted into two standard forms:

$$AX^2 + BY^2 + CZ^2 + J = 0 \quad \text{or} \quad AX^2 + BY^2 + IZ = 0.$$

Ellipsoid: $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$

Elliptic Paraboloid: $\frac{z}{c} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

Hyperbolic Paraboloid: $\frac{z}{c} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$

Cone: $\frac{z^2}{c^2} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

Hyperboloid of one sheet: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$

Hyperboloid of two sheets: $-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$

Chapter 13

P1

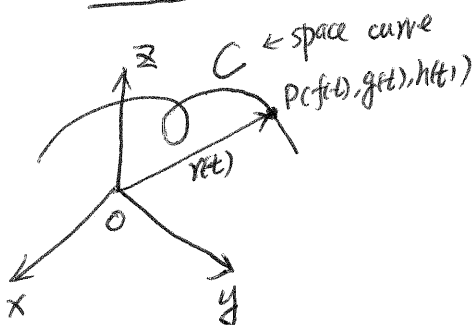
§13.1.

Vector function: $\gamma(t) = \langle f(t), g(t), h(t) \rangle$

f, g, h component functions of γ .

$$\lim_{t \rightarrow a} \gamma(t) = \langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \rangle$$

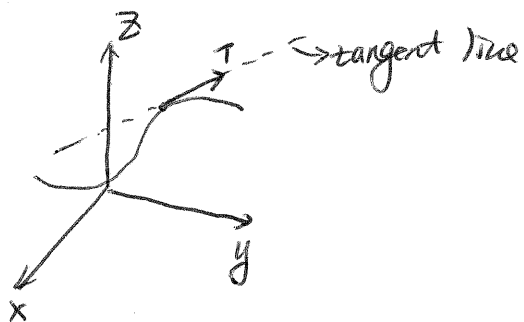
γ is continuous at a if $\lim_{t \rightarrow a} \gamma(t) = \gamma(a)$



$$x = f(t), \quad y = g(t), \quad z = h(t)$$

$\hookrightarrow$ parametric equations of the space curve C .

§13.2.



$\gamma'(t) \rightarrow$ tangent vector
 $T(t) = \frac{\gamma'(t)}{|\gamma'(t)|}$ called unit tangent vector.

Differentiation Rules

$$\frac{d}{dt} [u(t) \pm v(t)] = u'(t) \pm v'(t)$$

$$\frac{d}{dt} [\alpha u(t)] = \alpha u'(t)$$

$$\frac{d}{dt} [f(t)u(t)] = f'(t)u(t) + f(t)u'(t)$$

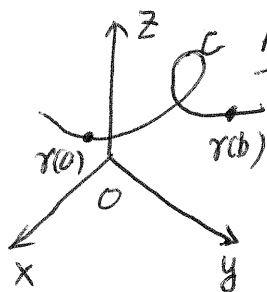
$$\frac{d}{dt} [u(t) \cdot v(t)] = u'(t) \cdot v(t) + u(t) \cdot v'(t)$$

$$\frac{d}{dt} [u(t) \times v(t)] = u'(t) \times v(t) + u(t) \times v'(t)$$

$$\frac{d}{dt} [u(f(t))] = f'(t)u'(f(t)) \rightarrow \text{Chain Rule.}$$

Theorem: If $|\gamma(t)| = C$ (a constant), then $\gamma(t) \cdot \gamma'(t) = 0$, i.e. $\gamma(t)$ is orthogonal to $\gamma'(t)$.

Integrals Let $R(t) = \int \gamma(t) dt = \langle \int f(t) dt, \int g(t) dt, \int h(t) dt \rangle$, then
 $\int_a^b \gamma(t) dt = R(b) - R(a)$.



Arc Length of the curve C from $r(a)$ to $r(b)$:

$$L = \int_a^b |r'(t)| dt$$

$$= \int_a^b \sqrt{(f'(t))^2 + (g'(t))^2 + (h'(t))^2} dt$$

Arc length function: the arc length from $r(a)$ to $r(t)$

$$s(t) = \int_a^t |r'(u)| du \rightarrow \frac{ds}{dt} = |r'(t)|$$

$\rightarrow$ one-to-one function.

Curvature of a curve

$$k = \left| \frac{dT}{ds} \right| = \frac{|T'(t)|}{|r'(t)|} \rightarrow \text{formula 1}$$

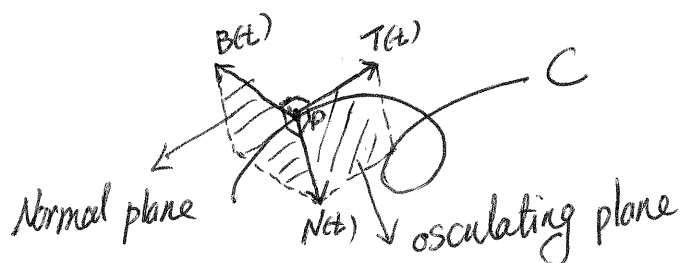
(at a general point)

or

$$k = \frac{|r'(t) \times r''(t)|}{|r'(t)|^3} \rightarrow \text{formula 2}$$

Principal unit vector

$$N(t) = \frac{T'(t)}{|T'(t)|}$$



Binormal vector

$$B(t) = T(t) \times N(t)$$

§ 13.4.

Velocity: $v(t) = r'(t)$ where $r(t)$ is the position vector in $\mathbb{R}^3$.

$$\text{speed} = |v(t)| = |r'(t)|$$

Acceleration: $a(t) = v'(t) = r''(t)$.

$$\text{Thus } r(t) = \int v(t) dt + C_1, \quad a(t) = \int v(t) dt + C_2$$

where C_1, C_2 are determined by initial conditions.

Newton's Second Law: $F(t) = m a(t)$
 $\downarrow \quad \downarrow \quad \searrow$
 force mass acceleration.